

# Technology Stack

|               |                                                    |
|---------------|----------------------------------------------------|
| Date          | 1 October 2026                                     |
| Team ID       | SWTID-2026-4683                                    |
| Project Name  | EduGenie: Google Gemini Powered Learning Assistant |
| Maximum Marks | 2 Marks                                            |
| Submitted by  | Dechen                                             |

## Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

## Technology Stack Details

| S.No | Architecture Component / Layer                                | Technology Chosen | Justification / Purpose                                                                                                                                                 |
|------|---------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Frontend / Client-Side<br><br>(e.g. React, Angular, HTML/CSS) | HTML + CSS        | Used for the simple web interface. HTML provides the page structure and forms, while CSS provides responsive styling for buttons, input areas and the result container. |

| S.No | Architecture Component / Layer                                        | Technology Chosen                 | Justification / Purpose                                                                                                                                                                         |
|------|-----------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | Backend / Server-Side<br><br>(e.g. Node.js, Python/Django, Java)      | Python + FastAPI + Uvicorn        | FastAPI provides lightweight RESTful API endpoints for the EduGenie modules. Python supports the AI modules and integrations, while Uvicorn runs the application as an ASGI server.             |
| 3    | Database / Data Storage<br><br>(e.g. MySQL, MongoDB, PostgreSQL)      | No persistent database specified  | The project document does not specify a database or persistent data store. The documented workflow sends user input to the relevant module and returns the generated result.                    |
| 4    | Cloud / Hosting / Deployment<br><br>(e.g. AWS, Azure, Heroku, Docker) | Google Gemini API + Local Uvicorn | Gemini provides cloud-based AI inference for Q&A, quizzes, summaries and learning recommendations. The documented application can also be run locally with Uvicorn for development and testing. |
| 5    | Version Control & CI/CD<br><br>(e.g. GitHub, GitLab, Jenkins)         | Git / GitHub                      | The project is maintained in a GitHub repository, supporting source-code versioning, collaboration and organized project development.                                                           |

| S.No | Architecture Component / Layer                                           | Technology Chosen                     | Justification / Purpose                                                                                                                                                                                        |
|------|--------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6    | Third-Party APIs / Other Tools<br><br>(e.g. Stripe, Twilio, Google Maps) | Google Gemini API<br>Google AI Studio | The Gemini API supplies generative AI capabilities used by EduGenie's Q&A;, quiz generation, summarization and personalized learning-path features. Google AI Studio is used to create and manage the API key. |
| 7    | AI / ML Model                                                            | Gemini 1.5 Pro<br>LaMini-Flan-T5-783M | Gemini 1.5 Pro is used for Q&A;, summarization, quiz generation and learning paths. LaMini-Flan-T5-783M is used for lightweight concept explanation and is described as CPU-compatible.                        |
| 8    | Frontend Templating                                                      | Jinja2                                | Jinja2 is used as the HTML templating engine for the documented templates directory and supports rendering the web interface.                                                                                  |
| 9    | Core Python Libraries / Dependencies                                     | Python 3.10+<br>requirements.txt      | Python 3.10+ is the documented prerequisite, while requirements.txt manages the project's Python dependencies needed to run the application.                                                                   |

## Technology Stack Summary

Frontend: HTML/CSS with Jinja2 templating. Backend: Python, FastAPI and Uvicorn. AI: Gemini 1.5 Pro and LaMini-Flan-T5-783M. Third-party service: Google Gemini API / Google AI Studio. Deployment: documented local execution through Uvicorn. Storage: no persistent database is specified in the project document.

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